

UI Design System Specification v0.1

Normative visual and component baseline for the CS-AML MVP analyst interface

Design-system axiom

The visual system SHALL communicate analytical state, provenance, uncertainty, permission, review status, and operational risk without implying guilt, suspicion, certainty, or importance merely through visual prominence.

Document control	Value
Document ID	CSAML-UI-DS-0.1
Status	Normative UI design-system baseline for MVP 0.1
Depends on	UX Specification; Information Architecture Specification; Screen Inventory; Wireframe Specification
Primary consumers	Product design, frontend engineering, QA, accessibility, product owner
Baseline theme	Light, calm, data-dense, evidence-first
MVP frontend reference	React + TypeScript + Tailwind CSS 4

1. Purpose, Scope, and Ownership

This specification defines the shared visual language and reusable UI component system for the CS-AML application. It standardises tokens, typography, spacing, color semantics, component anatomy, component states, data-density rules, analytical visual semantics, accessibility rules, and frontend implementation conventions.

1.1 What this document owns

- Design tokens: color, typography, spacing, size, border, radius, elevation, motion, opacity, and focus.
- Reusable UI components and their visual/semantic states.
- Analytical visualization semantics for graph, timeline, value flow, confidence, provenance, and uncertainty.
- Accessibility constraints and component-level interaction affordances where visual styling is involved.
- Frontend design-system implementation and governance conventions.

1.2 What this document does not own

Concern	Owning specification
Task flows, interaction behaviour, feedback, safety	CS-AML UX Specification
Navigation, taxonomy, object hierarchy, findability	CS-AML Information Architecture Specification
Screen catalogue and canonical screen IDs	CS-AML Screen Inventory
Screen regions and low-fidelity composition	CS-AML Wireframe Specification
Functional behaviour and acceptance	PRD / SRS / MVP Engineering Breakdown

Non-overlap rule

A design-system component MAY provide styles and states for navigation, forms, reviews, or graph views, but SHALL NOT redefine the information hierarchy, task flow, or business rule of those features.

2. Design Principles

ID	Principle	Normative intent
DS-P01	Evidence first	The visual hierarchy SHALL make provenance and supporting evidence easy to inspect without making allegations visually dominant.
DS-P02	Uncertainty visible	Unknown, disputed, inferred, reconstructed, hypothetical, and verified states SHALL be perceptibly different.

ID	Principle	Normative intent
DS-P03	Calm over alarm	The baseline UI SHALL avoid threat-dashboard aesthetics, excessive red, flashing indicators, or dramatic risk-score presentation.
DS-P04	No guilt by color	Entity type, centrality, case participation, or network prominence SHALL NOT be encoded as suspicion or guilt.
DS-P05	Accessible by default	Meaning SHALL NOT depend on color alone; keyboard focus, contrast, labels, and readable density are mandatory.
DS-P06	Dense but legible	The system SHOULD support professional investigative density while preserving grouping, scanability, and touch/click target adequacy.
DS-P07	Canonical consistency	The same status or analytical concept SHALL use the same token and component semantics across all screens.
DS-P08	Reversible state awareness	Merge, review, supersession, conflict, and destructive actions SHALL have visual cues that reveal reversibility or permanence.
DS-P09	Security is visible when useful	Classification, restricted handling, and protected-source states SHALL be visible to authorised users without unnecessarily exposing sensitive details.
DS-P10	Print/export parity	Critical semantic distinctions SHALL survive screenshots, grayscale print, PDF export, and copied report visuals.

3. Design Token Architecture

All visual values used by production components SHOULD be expressed through semantic design tokens rather than one-off literal styling. Component code SHALL prefer semantic tokens over raw palette values.

3.1 Token layers

Layer	Purpose	Examples
Primitive	Raw visual values	blue.700, gray.100, space.4, radius.md
Semantic	Meaningful UI role	surface.canvas, text.muted, border.default, status.warning
Analytical	CS-AML analytical meaning	flow.documented, proposition.inferred, confidence.low
Component	Local component mapping	button.primary.bg, badge.disputed.bg, graph.edge.reconstructed

3.2 Naming rules

- Use lowercase dot notation in implementation tokens, e.g. ``color.text.primary``.
- Do not encode a specific hue into semantic token names such as ``status.error``; the hue may change without changing meaning.
- Analytical tokens SHALL use domain meaning, not emotional labels such as “dangerous entity”.
- Every token intended to communicate state SHALL define light-theme contrast requirements and a non-color companion cue where material.

3.3 Core token requirements

ID	Requirement
DS-001	Production UI SHALL source colors, spacing, typography, radius, elevation, and state styling from versioned design tokens.
DS-002	Semantic state tokens SHALL NOT be hard-coded independently in feature modules.
DS-003	Analytical state tokens SHALL map consistently across graph, timeline, tables, badges, forms, and export-ready views.
DS-004	Token changes that alter analytical meaning SHALL be reviewed by UX and domain owners, not frontend engineering alone.

4. Color System

The MVP baseline uses a restrained light theme. Color is used to establish hierarchy and state, not to create a “security operations” aesthetic. Red is reserved primarily for destructive/error conditions; it is not a generic suspicion color.

4.1 Primitive palette baseline

Token	Hex	Usage	Swatch
Navy 900	#17324D	Primary headings, strong chrome	
Blue 700	#285C8E	Primary action/accent	
Ink 900	#16212B	Primary text	
Slate 600	#5E6B75	Secondary text	
Line 300	#D7DEE5	Borders/dividers	
Canvas	#F7F9FB	Application background	
Green 700	#246B4A	Confirmed success/completed	
Amber 700	#8A5A00	Warning/review needed	
Red 700	#A13333	Error/destructive action	
Purple 700	#5B4B8A	Hypothesis/analytical construct	
Cyan 700	#236A73	Documented/derived informational state	

4.2 Semantic color roles

Semantic token	Meaning	Do not use for
color.status.success	Completed/confirmed operational success	“Person is clean” or innocence claim
color.status.warning	Needs attention, incomplete, review required	Suspicion
color.status.error	System error, invalid state, destructive action	Adverse assessment by default
color.status.info	Informational/system context	Evidence strength
color.classification.restricted	Restricted handling visibility	Analytical guilt
color.analytical.construct	Hypothesis/inference construct	Verified fact

Critical prohibition

A person, organisation, company, address, wallet, or other entity SHALL NOT become red merely because it is a subject, appears in an investigation, has many graph links, matches a typology, or is associated with an adverse hypothesis.

5. Typography

Aptos is the reference documentation font; the product UI SHOULD use a widely available, highly legible system sans stack or an approved equivalent. The design system is font-family-neutral at the semantic level.

Token	Reference size	Weight	Use
type.display	28-32 px	600	Rare product-level titles
type.h1	24 px	600	Screen title
type.h2	20 px	600	Major section
type.h3	16 px	600	Panel/subsection title
type.body	14-16 px	400	Primary content
type.body.compact	13-14 px	400	Dense tables and metadata
type.label	12-13 px	500	Form labels, chips
type.caption	11-12 px	400	Secondary provenance and helper text
type.mono	12-14 px	400	IDs, hashes, technical identifiers

- Body text SHOULD normally render at 14 px or larger in the product UI.
- Critical evidence excerpts, assessments, and review text SHOULD use at least 14 px with comfortable line height.
- Monospace MAY be used for hashes, registration IDs, URLs, wallet addresses, and machine identifiers, but not for long prose.
- All-caps SHALL NOT be used for long status labels; reserve it for short controlled labels where scanability improves.

6. Spacing, Density, and Layout Tokens

Token	Value	Typical use
space.1	4 px	Tight icon/text gap
space.2	8 px	Inline groups

Token	Value	Typical use
space.3	12 px	Compact controls
space.4	16 px	Default component padding
space.5	20 px	Panel content
space.6	24 px	Section spacing
space.8	32 px	Major section separation
space.12	48 px	Large layout separation

6.1 Density modes

Mode	Purpose	Constraints
Comfortable	Reading, review, authoring	Larger vertical rhythm; default for prose-heavy views
Compact	Registers, evidence lists, entity tables	Reduced vertical padding; text remains ≥ 13 px; targets remain keyboard accessible
Analytical canvas	Graph/timeline/value flow	Maximise working area while keeping inspector/legend readable
Presentation/export	Screenshots, PDF, referral	Semantic distinction remains visible without hover or interaction

The UI MAY allow users to choose comfortable versus compact density for data-heavy surfaces. A density preference SHALL NOT hide provenance, confidence, classification, or uncertainty indicators.

7. Borders, Radius, Elevation, and Surfaces

Token	Baseline	Rule
radius.sm	4 px	Inputs, chips, compact controls
radius.md	8 px	Cards, panels, menus
radius.lg	12 px	Large dialogs/sheets only
border.default	1 px	Neutral structural separation
elevation.1	Subtle shadow	Floating menus/tooltips
elevation.2	Moderate shadow	Modal/dialog only
surface.canvas	Very light neutral	App background

Token	Baseline	Rule
surface.panel	White	Primary workspace panels
surface.subtle	Light neutral	Grouped metadata/read-only regions

Persistent cards SHOULD rely more on borders and spacing than on heavy shadows. Excessive elevation is discouraged because the interface is a professional evidence workspace, not a consumer dashboard.

8. Iconography

- Icons SHALL have a text label or accessible name when they invoke actions.
- Icon shape SHALL NOT be the only carrier of analytical meaning where confusion is plausible.
- Use a single icon family with consistent stroke weight.
- Avoid law-enforcement symbolism (handcuffs, targets, crosshairs) as default analytical icons.
- Entity-type icons MAY distinguish person, organisation, company, asset, account, wallet, address, contract, and project, but SHALL NOT imply adverse status.

9. Interactive State Model

State	Visual requirements	Accessibility
Default	Normal surface/border/text	Normal accessible name
Hover	Subtle surface/border change	Never only indication of availability
Focus	High-contrast focus ring	Visible for keyboard focus; not removed
Active/selected	Persistent border/fill + selected indicator	aria-selected/pressed where applicable
Disabled	Reduced emphasis but readable	Disabled reason available when useful
Loading	Skeleton/progress without layout jump	Announce long-running operations
Error	Error text + icon + field relationship	Do not rely on red alone
Success	Confirmation text/icon	Do not auto-dismiss critical confirmation too fast
Conflict	Distinct conflict banner/region	Describe competing version and recovery
Read-only	Read-only label and neutral surface	Avoid disabled styling if content remains inspectable

ID	Requirement
DS-005	Keyboard focus SHALL be visually obvious on every actionable component.
DS-006	Error, warning, success, and analytical states SHALL combine color with text, icon, pattern, border, or label.
DS-007	Disabled high-impact actions SHOULD explain the prerequisite when the user is authorised to know it.
DS-008	Loading and async states SHALL preserve enough context that investigators know which object/action is pending.

10. Analytical State Semantics

Analytical state is domain meaning, not generic UI status. These states require dedicated tokens and consistent rendering across badges, tables, detail pages, graph, timeline, value flow, and exported visuals.

10.1 Proposition state

State	Visual treatment	Meaning
Claim	Outlined neutral badge + “Claim” label	Statement attributed to a source; not verified fact
Fact	Solid neutral/blue badge + “Fact” label	Verified/time-bounded proposition per methodology
Inference	Purple outlined badge + “Inference”	Analytical interpretation
Disputed	Amber patterned/outlined badge	Material contradiction or challenge exists
Unknown	Gray dashed treatment + “Unknown”	Information not established
Candidate	Light outlined badge + “Candidate”	Suggested entity/match/relationship awaiting review

10.2 Confidence

Level	Treatment	Constraint
High	Label + strong border	Never represented as certainty/proof
Moderate	Label + medium border	Basis must remain inspectable
Low	Label + light/dashed border	Do not hide as secondary metadata
Insufficient basis	Neutral warning-style label	Must not look like adverse result

Confidence rule

Confidence is confidence in an assessment, not a probability of guilt. The UI SHALL NOT convert confidence language into a “risk meter”, percentage gauge, or threat score unless a separately validated model explicitly requires it.

11. Value-Flow Visual Semantics

Class	Line token	Mandatory text cue	Meaning
DIRECT	Solid line	DIRECT label	Direct transaction/value

Class	Line token	Mandatory text cue	Meaning
			evidence where authoritative evidence supports it
DOCUMENTED	Solid/double-accent line	DOCUMENTED label	Documented value relationship such as contract, award, purchase
RECONSTRUCTED	Dashed line	RECONSTRUCTED label	Analyst reconstruction from multiple records
HYPOTHETICAL	Dotted line	HYPOTHETICAL label	Analytical possibility/hypothesis

- Color MAY supplement but SHALL NOT replace line style and text label.
- Legends SHALL remain visible or easily accessible in analytical workspaces and exports.
- Unknown values SHALL display “Unknown”, not 0.
- Ranges and approximate values SHALL retain approximation markers.
- A chain containing mixed epistemic classes SHALL preserve the class of each leg; the whole chain SHALL NOT inherit the strongest class.

12. Investigation Graph Design Rules

Element	Design rule
Node type	Use neutral type-specific icon/shape; do not encode suspicion.
Node size	Default size SHOULD be stable. Centrality MAY be optional analytic overlay, not default guilt cue.
Edge	Type + status + evidence availability must be inspectable.
Inferred edge	Use distinct analytical styling and label.
Selection	High-contrast outline; do not change semantic status.
Restricted object	Render only according to permission policy; never reveal hidden node count if prohibited.
Provenance	Inspector SHALL expose relationship record and supporting evidence.
Community colors	Exploratory palette only; legend states algorithmic grouping, not culpability.

Graph prohibition

Red nodes, skull icons, target rings, heat-map threat coloring, or automatic “high risk” styling SHALL NOT be assigned solely from degree, centrality, path proximity, typology match, or presence in a case.

13. Timeline Design Rules

- Exact, approximate, month-only, year-only, range, and unknown date precision SHALL be visually distinguishable.
- Events SHALL expose type, involved objects, date precision, and evidence/provenance.
- Current versus historical relationship state SHALL be distinguishable without requiring hover.
- Dense timelines MAY collapse low-priority items, but hidden count and filtering logic SHALL be visible.
- Timeline color SHOULD primarily encode event category or selection, not adverse status.

14. Component Catalogue

ID	Component	Primary use
DS-C001	Button	Primary, secondary, tertiary, danger, icon
DS-C002	Text input	Text, URL, identifier, amount
DS-C003	Select / combobox	Controlled vocabulary, entity picker
DS-C004	Date / date precision	Exact/approximate/range/unknown
DS-C005	Checkbox / radio / switch	Explicit state controls
DS-C006	Badge / status pill	Lifecycle, analytical state, classification
DS-C007	Chip / token	Filters, entity/relationship tags
DS-C008	Alert / callout	Error, warning, info, integrity
DS-C009	Toast	Transient non-critical feedback
DS-C010	Table / data grid	Registers and dense object lists
DS-C011	Card / panel	Grouped object/detail region
DS-C012	Tabs	Local object/workspace sections
DS-C013	Breadcrumb / context trail	Context orientation; IA labels authoritative
DS-C014	Drawer / side inspector	Evidence/provenance/object detail
DS-C015	Modal / dialog	High-focus bounded actions
DS-C016	Confirmation dialog	Destructive/high-impact confirmation
DS-C017	Evidence citation	Source/evidence ID + precise locator
DS-C018	Provenance trail	Backward lineage chain
DS-C019	Confidence badge	Assessment confidence label + rationale entry
DS-C020	Flow class badge	Four mandatory value-flow classes

ID	Component	Primary use
DS-C021	Entity identity block	Type, canonical name, aliases/status
DS-C022	Relationship summary	Type, endpoints, dates, evidence/confidence
DS-C023	Review decision panel	Approve/request changes/reject + rationale
DS-C024	Classification banner	Handling/classification context
DS-C025	Empty state	Explain why empty and next permitted action
DS-C026	Permission state	Non-disclosing denial / access request guidance
DS-C027	Integrity warning	Hash mismatch/evidence integrity issue
DS-C028	Version/supersession banner	Current/superseded/retracted state
DS-C029	Skeleton / progress	Async loading
DS-C030	Legend	Analytical semantics/key

15. Actions and Buttons

Variant	Use	Rule
Primary	Single main task in current region	Generally one primary action per major panel
Secondary	Alternative non-destructive action	May coexist with primary
Tertiary	Low-emphasis utility action	Text/ghost style
Danger	Destructive/irreversible action	Red semantic reserved for destructive actions; confirmation often required
High-impact approval	Approve dissemination/review	Visually explicit; label names consequence, e.g. “Approve for dissemination”

- Ambiguous labels such as “OK” SHALL NOT be used for consequential actions.
- Buttons that create irreversible or externally visible effects SHALL describe the effect.
- Primary button styling SHALL NOT be used to nudge users toward an adverse analytical conclusion.
- Approval and rejection controls SHOULD be visually balanced where human judgement is required.

16. Forms and Data Entry Components

Rule	Requirement
Labels	Persistent labels above/adjacent to controls; placeholder is not a label.
Required fields	Mark explicitly; do not rely on color.
Helper text	Use for methodology or format constraints; keep concise.
Validation	Inline, specific, and linked to the field; retain user input.
Unknown values	Provide explicit Unknown / Not assessed option where domain model allows it.
Confidence/status	Use controlled values plus rationale field where required.
Entity references	Show object type + canonical name + stable ID/context disambiguator.
Evidence links	Use searchable/picker components with provenance preview.
Long analysis	Use autosave/version-aware editor with visible save/conflict state.

17. Data Tables and Registers

Tables are a primary professional workspace and SHALL support dense information without losing state semantics or accessibility.

- Header row is visually distinct and programmatically identified.
- Column widths follow content importance; narrative columns receive more space than short statuses.
- Sorting state is explicit; default sort is documented per screen.
- Filters appear as removable chips/tokens and reflect current query scope.
- Row selection is visually distinct from analytical status.
- Bulk actions SHALL be permission-aware and avoid accidental mass changes.
- Long identifiers MAY truncate visually only if copy/full-value access remains available.
- Critical status or classification SHALL not disappear when horizontal scrolling is required.

ID	Requirement
DS-009	Data-grid selection styling SHALL NOT reuse adverse analytical status colors.
DS-010	Tables SHALL expose empty, loading, filtered-empty, error, and partial-data states.

ID	Requirement
DS-011	Restricted rows SHALL be omitted or represented only according to permission policy; UI SHALL NOT leak existence through pagination/counts when policy prohibits it.

18. Badges, Labels, Alerts, and Banners

18.1 Badge classes

Class	Examples	Visual intent
Lifecycle	Draft, Active, Closed, Superseded	Operational state
Analytical	Claim, Fact, Inference, Disputed, Unknown	Epistemic state
Confidence	High, Moderate, Low, Insufficient basis	Assessment confidence
Classification	Public, Internal, Sensitive, Restricted, Source-Protected	Handling requirement
Review	Needs review, Changes requested, Approved	Workflow state
Flow	Direct, Documented, Reconstructed, Hypothetical	Value-flow epistemics

18.2 Banner hierarchy

- Persistent banners are appropriate for classification, supersession, integrity failure, read-only state, and significant access restrictions.
- Warnings SHALL describe the condition and available action; avoid generic “Danger” wording.
- Integrity warnings outrank convenience notices and SHOULD remain until resolved or acknowledged per policy.
- Source-protection banners SHALL avoid unnecessarily naming the protected source.

19. Drawers, Inspectors, Modals, and Dialogs

Pattern	Preferred use	Avoid
Side inspector	Provenance, evidence, node/edge details while preserving analytical context	Deep multi-step editing
Drawer	Secondary workflow with context retained	Critical approval if user cannot review full context

Pattern	Preferred use	Avoid
Modal	Short bounded decision/input	Long reports or complex evidence analysis
Confirmation dialog	Destructive or externally consequential action	Routine saves
Full-screen workspace	Graph, value flow, review when spatial context is primary	Simple record editing

20. Evidence and Provenance Components

Evidence/provenance components are first-class design-system elements, not feature-specific decorations.

Component	Mandatory visible elements
Evidence citation	Evidence ID or concise label; source; precise location/page/region; open action
Original/derivative marker	Original vs OCR/translation/extract/processed; parent link
Integrity block	Hash status; acquisition timestamp; verification state where relevant
Provenance trail	Source → Evidence → Extract → Fact/Indicator → Assessment links as permitted
Reliability/credibility	Separate source reliability and information credibility controls
Protected-source reference	Safe alias/opaque reference; no unnecessary identity exposure

Evidence visibility rule

The system **SHOULD** make it easier to inspect evidence than to copy an unsupported allegation. Analytical statements without required provenance should appear incomplete, not polished.

21. Review and Dissemination Components

- Review panel **SHALL** show product version, author, reviewer, classification, unresolved comments, and decision state.
- Approval controls **SHALL** not visually privilege approval over request-changes/reject where policy requires independent judgement.
- External dissemination screens **SHALL** expose recipient, purpose, version, classification, minimisation/redaction state, and restrictions before final confirmation.

- Export preview SHALL visually distinguish included, excluded, and redacted material.
- Sharing-log entries are read-only after finalization for ordinary analyst roles.

ID	Requirement
DS-012	High-impact approval components SHALL state the consequence in the action label.
DS-013	Export/dissemination confirmation SHALL display the exact product version and intended recipient/purpose.
DS-014	Redacted or excluded material SHALL be perceptibly different in preview without exposing protected content.

22. Classification and Security Visuals

Classification	Reference treatment	Notes
PUBLIC	Neutral/low emphasis	No implication of low evidentiary value
INTERNAL	Blue-gray label	Internal handling
SENSITIVE	Amber-outline label	Attention to handling, not suspicion
RESTRICTED	Strong amber/brown label + lock icon/text	Need-to-know handling
SOURCE-PROTECTED	Distinct shield/lock label; minimal detail	Identity compartmentalisation; avoid source name

- Classification styling SHALL represent handling requirements, not analytical severity.
- Protected-source state SHALL not be indexed into general-purpose visual components if that leaks existence.
- Permission-denied screens SHALL use non-disclosing copy where policy requires concealment of object existence.

23. Responsive and Viewport Baseline

CS-AML is desktop-first because graph, evidence review, and analytical comparison require spatial context. Responsive behaviour must preserve critical tasks rather than merely shrink the desktop layout.

Viewport	Baseline behaviour
>=1440 px	Full multi-panel analytical workspace; optional persistent inspector
1024-1439 px	Two-panel workspace; inspector may overlay/drawer

Viewport	Baseline behaviour
768-1023 px	Stacked panels; analytical canvas remains usable with explicit panel switching
<768 px	Limited/mobile support for review, lookup, simple tasks; complex graph/value-flow authoring MAY be unsupported in MVP

Mobile scope

MVP responsive support does not require parity for every analytical authoring task. Unsupported complex tasks SHALL fail gracefully with a clear explanation rather than present a misleadingly simplified workflow.

24. Accessibility Baseline

The design system SHALL target WCAG 2.2 AA-equivalent accessibility for product UI wherever applicable, with special attention to keyboard use, focus, contrast, color independence, dense tables, graph alternatives, and error recovery.

ID	Requirement
DS-015	Normal text and essential UI controls SHALL meet applicable contrast targets.
DS-016	Every interactive component SHALL be operable by keyboard unless the interaction is inherently spatial, in which case an equivalent accessible path SHALL exist where required.
DS-017	Focus order SHALL follow task order, not visual implementation accident.
DS-018	Graph, timeline, and value-flow views SHALL provide textual/list/table alternatives for material analytical content.
DS-019	Color SHALL NOT be the sole means of conveying value-flow class, confidence, error, classification, or analytical state.
DS-020	Motion SHALL respect reduced-motion preferences and SHALL NOT be required to perceive state changes.

24.1 Minimum target sizes

- Primary pointer targets SHOULD be approximately 40×40 CSS px or larger where practical.
- Compact data-grid controls MAY be smaller when keyboard access and sufficient spacing are preserved.

- Icons used alone as controls require accessible names and tooltips/help where meaning is not universally clear.

25. Motion and Transition

- Motion is functional, not decorative.
- Use short transitions for panel expansion, selection, and context preservation.
- Avoid pulsing/flashing “risk” indicators.
- Async completion **SHOULD** use stable confirmation and not rely only on transient animation.
- Reduced-motion mode **SHALL** remove non-essential animation.

26. Frontend Implementation Conventions

The reference MVP frontend is React + TypeScript + Tailwind CSS 4. The design system SHOULD expose semantic tokens as CSS custom properties and reusable typed components. Raw Tailwind utility combinations SHOULD NOT become de facto component definitions scattered across features.

26.1 Reference structure

```
frontend/  
  src/  
    design-system/  
      tokens/  
      components/  
      patterns/  
      icons/  
      styles/  
      tests/  
      index.ts  
    features/  
      cases/  
      evidence/  
      entities/  
      analysis/  
      products/
```

26.2 CSS variable examples

```
--color-text-primary: #16212B;  
--color-surface-canvas: #F7F9FB;  
--color-border-default: #D7DEE5;  
--color-action-primary: #285C8E;  
--color-state-error: #A13333;  
--flow-direct-style: solid;  
--flow-reconstructed-style: dashed;
```

ID	Requirement
DS-021	Feature modules SHOULD consume design-system components before creating local equivalents.
DS-022	Design-system components SHALL expose typed variants for analytical/status semantics rather than arbitrary color props.
DS-023	Raw hex values SHOULD NOT appear in feature-level production code except documented exceptional cases.
DS-024	Component accessibility tests SHALL be included in CI for critical components.

ID	Requirement
DS-025	Breaking component/token changes SHALL follow versioning and migration notes.

27. Component API and Composition Rules

- Prefer semantic props: ``state="disputed"`, `flowClass="reconstructed"`, `classification="restricted"`` rather than `color="orange"``.`
- Components SHOULD expose slots/composition points for provenance and metadata without requiring feature forks.
- Critical components SHALL support loading, error, empty, read-only, restricted, and disabled states where applicable.
- Component defaults SHALL be safe: destructive variants are opt-in; unrestricted data exposure is never a visual default.
- UI copy belongs to product/feature layer except universal design-system labels such as generic loading/error affordances.

28. Design-System Verification

Test class	Minimum verification
Visual regression	Critical components and analytical states at supported themes/viewports
Accessibility	Keyboard, focus, accessible name, semantics, contrast, color independence
State coverage	Default/hover/focus/disabled/loading/error/read-only/conflict where applicable
Responsive	Critical components at desktop/tablet/mobile baseline
Export parity	Value-flow classes and analytical labels survive PDF/screenshot/grayscale checks
Semantic consistency	Same domain state maps to same token/component across features
Security leakage	Restricted/protected states do not expose hidden labels or counts in shared components

28.1 Component Definition of Done

- Semantic purpose and variants documented.
- Accessible name/role/state behaviour defined.
- Keyboard/focus verified.
- Critical states implemented.
- No feature-specific raw color dependency.

- Visual regression baseline stored.
- Usage example available.
- Relevant analytical/provenance semantics tested.
- No contradiction with UX/IA/wireframe specifications.

29. Design-System Governance

The design system is a shared product asset. Visual changes that alter analytical meaning or safety require domain review; purely cosmetic changes still require accessibility and consistency review.

Change type	Required review
New primitive token	Design-system owner
New semantic token	Design-system owner + UX
New analytical token/state	UX + domain/methodology owner + frontend
New reusable component	UX/design + frontend + accessibility/QA
Breaking visual semantic change	Product + UX + domain + QA; migration note
One-off feature override	Discouraged; justification required

29.1 Versioning

- v0.x may evolve while MVP is under active design, but analytical semantics require explicit change records.
- Removed tokens/components SHALL have a migration path where production usage exists.
- Component and token changelog SHOULD be maintained in repository documentation.
- Screen-specific visual exceptions SHOULD be documented rather than silently diverging.

30. Traceability to CS-AML Product Surfaces

Design-system domain	Primary dependent screens/patterns
Forms/actions	Case charter, source/evidence registration, entity/relationship editing, assessment
Data grids	Cases, entities, evidence, products, sharing log, admin registers
Evidence/provenance	Evidence Reader, Entity Detail, Assessment, Product Editor, Review
Analytical semantics	Value Flow Workspace, Timeline, Hypothesis Matrix, Investigation Graph
Review/dissemination	Peer Review Workspace, Dissemination Approval, Export Builder
Security/classification	All canonical detail screens, source-protected flows,

Design-system domain	Primary dependent screens/patterns
	exports
System states	Access denied, edit conflict, integrity warning, async processing

Annex A — MVP Semantic Token Baseline

Token	Baseline	Meaning
color.text.primary	#16212B	Primary readable content
color.text.secondary	#5E6B75	Secondary metadata
color.surface.canvas	#F7F9FB	Application canvas
color.surface.panel	FFFFFF	Panel/card surface
color.border.default	#D7DEE5	Default border
color.action.primary	#285C8E	Primary action
color.state.success	#246B4A	Operational success
color.state.warning	#8A5A00	Attention/review
color.state.error	#A13333	Error/destructive
color.analytical.construct	#5B4B8A	Hypothesis/inference
color.analytical.documented	#236A73	Documented analytical state
focus.ring	#285C8E	Keyboard focus ring
space.base	4 px	Spacing scale base
radius.default	8 px	Default panels/controls

These values are a reference implementation baseline. Semantic meaning is normative; exact primitive hues MAY change if contrast, branding, or implementation requirements demand it, provided domain semantics remain stable.

Annex B — Analytical Legend Baseline

Concept	Required non-color cue	Recommended visual family
Claim	Text label “Claim”	Neutral outline
Fact	Text label “Fact”	Strong neutral/blue
Inference	Text label “Inference”	Purple outline
Disputed	Text label “Disputed” + icon/pattern	Amber
Unknown	Text label “Unknown” + dashed treatment	Gray
DIRECT flow	DIRECT label + solid line	Neutral/blue
DOCUMENTED flow	DOCUMENTED label + distinct solid line	Cyan
RECONSTRUCTED flow	RECONSTRUCTED label + dashed line	Purple/neutral

Concept	Required non-color cue	Recommended visual family
HYPOTHETICAL flow	HYPOTHETICAL label + dotted line	Gray/purple
Restricted classification	Text + lock symbol	Amber/brown
Source-Protected	Text + shield/lock symbol	Distinct secure treatment

Annex C — UI Design Review Checklist

- Does the screen use canonical design-system components where available?
- Are proposition, confidence, flow, classification, and review states semantically correct?
- Does any color, icon, size, ranking, or animation imply guilt or suspicion without methodological basis?
- Can critical meaning be understood without color?
- Are focus, keyboard, error, loading, empty, read-only, conflict, and permission states addressed?
- Are provenance and evidence links visually inspectable where decisions depend on them?
- Do graphs/timelines/value-flow views provide legends and accessible alternatives?
- Do destructive or external actions state their consequence?
- Are compact/dense layouts still readable and accessible?
- Will the visual semantics survive screenshot/PDF/grayscale export?
- Does the implementation use semantic tokens instead of feature-specific raw values?
- Has the change been checked against UX, IA, Screen Inventory, and Wireframe specifications?

Annex D — UI Design System Handoff Contract

Discipline	Owns	Consumes from UI DS
UX	Task flow, interaction behaviour, feedback, safety	Components, states, semantic visual tokens
Information Architecture	Navigation, hierarchy, taxonomy, labels	Component affordances and state presentation
Frontend Engineering	Implementation, component APIs, rendering	Tokens, component contracts, accessibility rules
QA / Accessibility	Verification	Expected variants/states and test requirements
Domain / Methodology	Analytical meaning	Visual mappings for epistemic/provenance states
Product	Scope/priorities	Reusable patterns and readiness evidence

Final release rule

UI completion requires correct semantic tokens/components, preserved analytical meaning, critical accessibility, required states, and consistency with the UX and IA specifications.